

Thunder is the acoustic shock wave caused by lightning. When a lightning bolt strikes, it superheats the surrounding air up to $(30,000^{\circ}\text{C})$ ($(54,000^{\circ}\text{F})$) in milliseconds. This extreme heat causes the air to expand explosively and then rapidly cool and contract, creating a massive sound wave that is heard as a clap or rumble. [[1](#), [2](#), [3](#), [4](#)]

The Science of Thunder

Because light travels significantly faster than sound (light takes just fractions of a second to reach the eye, while sound travels at about 343 meters per second), you always see lightning before you hear thunder. [[1](#)]

- Calculating Distance: Because sound travels roughly 1 kilometer in 3 seconds (or about 1 mile in 5 seconds), you can count the seconds between the flash of light and the crack of thunder to determine how far away the storm is.
- Rumbles vs. Claps: A sharp "clap" of thunder means the lightning struck very close to you. A deep, prolonged "rumble" is the sound arriving from different sections of a long lightning strike, echoing off clouds, hills, and the ground.
- Brontology: The scientific study of thunder is known as brontology, and the irrational fear of it is called brontophobia. [[1](#), [2](#), [3](#), [4](#), [5](#)]

For an in-depth breakdown of how atmospheric instability and extreme heat can lead to the sensation of constant, rolling thunder: